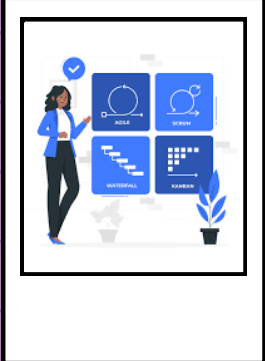


“PRESERVING ARCHITECTURAL STYLES IN THE SEARCH BASED DESIGN OF SOFTWARE PRODUCT LINE ARCHITECTURES

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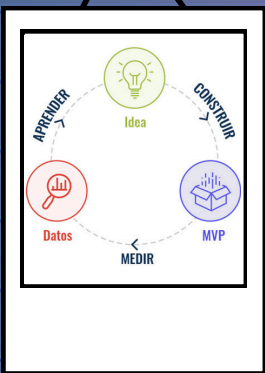
IMPORTANCE OF ARCHITECTURAL STYLES

Preserving architectural styles is crucial for maintaining consistency and quality in Software Product Line Architectures (SPLA).



SEARCH-BASED OPTIMIZATION

Search-Based Software Engineering (SBSE) techniques are used to explore and select optimal configurations within SPLA.



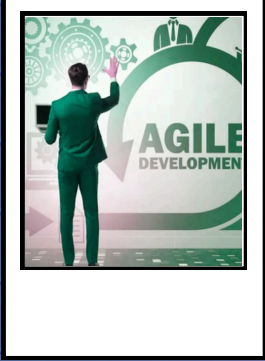
DESIGN CONSTRAINTS

The preservation of architectural styles imposes constraints that must be considered in the optimization process.



ARCHITECTURAL QUALITY

Maintaining proper architectural styles contributes to modularity, reusability, and software maintainability.



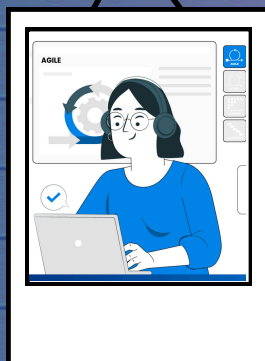
TRADE-OFFS IN OPTIMIZATION

There are trade-offs between adherence to architectural styles and other factors such as cost, performance, and flexibility.



PRESERVATION STRATEGIES

Mechanisms such as constraints in search algorithms can be implemented to ensure that architectural styles are not compromised.

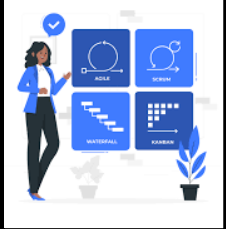


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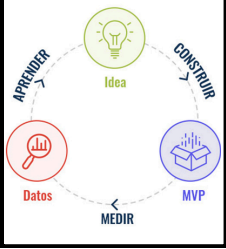
SPLA MODELING

The representation of SPLA must allow the inclusion of rules defining the allowed architectural styles.



IMPACT ON SOFTWARE EVOLUTION

Preserving architectural styles facilitates controlled evolution of the product line without affecting its core structure.



SOLUTION EVALUATION

Metrics and validation mechanisms must be defined to measure how well a solution adheres to established architectural styles.



APPLICATION OF EVOLUTIONARY ALGORITHMS

Methods such as genetic algorithms can be used to find optimal architectures that meet stylistic constraints.



AUTOMATION OF ARCHITECTURAL DESIGN

SBSE can automate the generation of architectures aligned with predefined design principles.



CHALLENGES AND FUTURE DIRECTIONS

Combining SBSE with architectural constraints still faces challenges, such as scalability and integration with software modeling tools.

